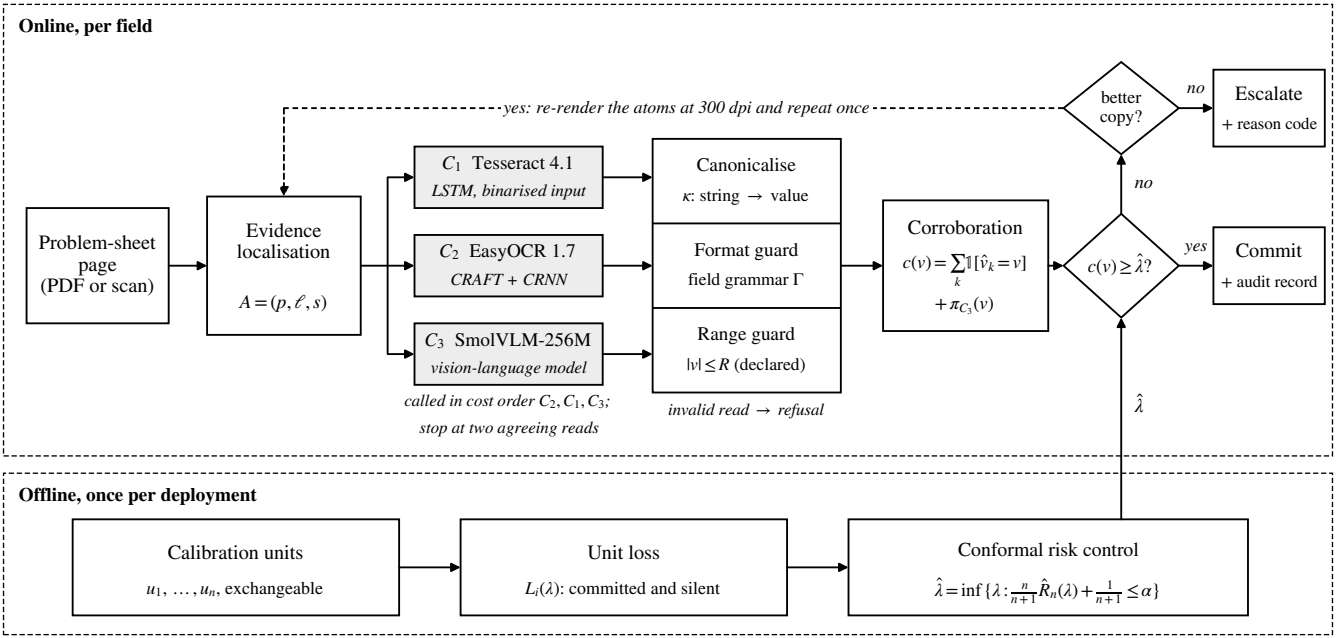
Online, per field
Problem-sheet page (PDF or scan)
Evidence localisation
$A = (p, \ell, s)$
C_1 Tesseract 4.1
LSTM, binarised input
C_2 EasyOCR 1.7
CRAFT + CRNN
C_3 SmolVLM-256M
vision-language model
called in cost order C_2, C_1, C_3; stop at two agreeing reads
Canonicalise
κ: string → value
Format guard
field grammar Γ
Range guard
$|v| \le R$ (declared)
invalid read → refusal
Corroboration
$c(v) = \sum_k \mathbb{1}[\hat{v}_k = v]$
$+ \pi_{C_3}(v)$
$c(v) \ge \hat{\lambda}$?
yes
Commit
+ audit record
no
better copy?
no
Escalate
+ reason code
yes: re-render the atoms at 300 dpi and repeat once
$\hat{\lambda}$
Offline, once per deployment
Calibration units
$u_1, \ldots, u_n$, exchangeable
Unit loss
$L_i(\lambda)$: committed and silent
Conformal risk control
$\hat{\lambda} = \inf\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\}$